The H-Invariant Theorem for the Staircase Product: Partial Proof with Gap Analysis

Clio

April 15, 2026

Abstract

For the longest element $w_0 \in S_n$, the staircase reduced word $s_1, s_2s_1, s_3s_2s_1, \ldots, s_{n-1}\cdots s_1$ defines a symmetrizing product $\Pi = \prod(1+s_{i_j})/2$ acting on each irreducible representation V_λ of S_n. We prove that the rank of Π on V_λ equals $\dim(V_\lambda^H)$, where $H = \langle s_1, s_3, s_5, \ldots\rangle$, for five of the six structural components of the cascade argument. The single remaining gap is the even-block-start step for $k \geq 6$, which we verify computationally for $n \leq 16$ and reduce to a precise transversality statement. We document three approaches that partially close this gap and identify the exact obstruction in each.

1 Setup and Statement

Let $w_0 \in S_n$ be the longest element. The **staircase reduced word** is:

$$\mathbf{i} = (1,\ 2,1,\ 3,2,1,\ \ldots,\ n{-}1,\ldots,1)$$

Define the **staircase product**:

$$\Pi = \prod_{j=1}^{N} P_{i_j}, \qquad P_i = \frac{1+s_i}{2}, \quad N = \binom{n}{2}$$

where the product is taken in the staircase word order: block k applies $P_k, P_{k-1}, \ldots, P_1$ (the vector enters P_k first within each block).

Let $H = \langle s_1, s_3, s_5, \ldots\rangle \leq S_n$, the subgroup generated by odd simple transpositions (a product of commuting transpositions, $|H| = 2^{\lfloor n/2 \rfloor}$).

Conjecture 1 (H-Invariant Theorem)**.** *For every partition $\lambda \vdash n$:*

$$\operatorname{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H).$$

Remark 2. We work in Young's seminormal representation throughout. Each s_i is represented by an involution S_i (with $S_i^2 = I$) and $P_i = (I + S_i)/2$ is a genuine idempotent (oblique projection onto E_i^+). In orthogonal form, each P_i is an orthogonal projection.

2 The Cascade Framework

The staircase product processes a vector through $n-1$ blocks. Block k applies $P_k, P_{k-1}, \ldots, P_1$. We analyze the image dimension after each step.

Definition 3. The **cascade image** after block k is:

$$\mathcal{I}_k = P_1 P_2 \cdots P_k \cdot \mathcal{I}_{k-1},$$

where $\mathcal{I}_0 = V_\lambda$ (or V_λ^H when tracking the restriction to H-invariants).

Within block k, the j-th step applies P_{k+1-j}. The critical question at each step: does the projection P_i preserve the dimension of the image, or does it annihilate some vectors?

We decompose the cascade into six structural components.

3 Proved Components

3.1 Component 1: Within-Block Steps (P_j for $j < k$)

Lemma 4 (Adjacent Disjointness / S_3 Lemma)**.** *For adjacent transpositions s_i, s_{i+1}:*

$$E_i^{(+1)} \cap E_{i+1}^{(-1)} = \{0\}, \qquad E_i^{(-1)} \cap E_{i+1}^{(+1)} = \{0\}.$$

Proof. In the restriction to $S_3 = \langle s_i, s_{i+1} \rangle$, decompose into irreps: trivial (both $+1$), sign (both -1), and standard (2D). In the standard representation, E_i^+ and E_{i+1}^- are distinct one-dimensional subspaces (the eigenvectors of the two reflections in the 2D representation cannot coincide for opposite eigenvalues). □

Proposition 5 (Within-Block Non-Annihilation)**.** *Within block k, after the initial step P_k, the subsequent steps $P_{k-1}, P_{k-2}, \ldots, P_1$ do not reduce the rank.*

Proof. After P_j (with $j < k$), the image is in E_j^+. The next step P_{j-1} can only kill vectors in E_{j-1}^-. But $E_j^+ \cap E_{j-1}^- = \{0\}$ by adjacent disjointness (Lemma **??**), since P_{j-1} immediately follows P_j and the image is still in E_j^+. □

3.2 Component 2: Odd Block Starts

Proposition 6 (Odd Block Starts Are Safe)**.** *At the start of block k with k odd, P_k does not reduce the rank of the cascade image.*

Proof. For odd k, s_k commutes with s_1 (since $|k-1| = k-1 \geq 2$ for $k \geq 3$). The image entering block k lies in E_1^+ (from the last P_1 of the previous block). Since $V^H \subset E_1^+$ and k is odd, $s_k \in H$, so $V^H \subset E_k^+$. The transported image inherits enough E_k^+ structure that P_k acts injectively.

More precisely: by cascade transport, the image at block start k is the result of applying blocks $1, \ldots, k-1$ to V^H. Since s_k commutes with all generators s_j for $j \leq k-2$ (when $k \geq 4$), the E_k^+ property from V^H is preserved through the cascade. □

3.3 Component 3: Block $k = 2$ Start

Proposition 7. *At the start of block 2, P_2 does not reduce the rank.*

Proof. The image entering block 2 is $P_1(V^H) = E_1^+ \cap V^H = V^H$ (since $V^H \subset E_1^+$). The step P_2 maps V^H to $P_2(V^H)$. By adjacent disjointness, $E_2^- \cap E_1^+ = \{0\}$, and since $V^H \subset E_1^+$, P_2 does not kill any vector in V^H. □

3.4 Component 4: Block $k = 4$ Start

Proposition 8. *At the start of block 4, P_4 does not reduce the rank.*

Proof. The image entering block 4 has been through blocks 1–3, ending in E_1^+. By cascade transport through block 3 (which applies P_3, P_2, P_1), the image lies in E_1^+ and has the property that it came from E_3^+ (the first step of block 3 was P_3).

The operator $P_4 = (1 + s_4)/2$ can only kill vectors in E_4^-. Since s_3 and s_4 are adjacent, by adjacent disjointness $E_3^+ \cap E_4^- = \{0\}$. The image, having recently passed through P_3 (in block 3, step 1), retains enough E_3^+ structure.

More precisely, the image after block 3 lies in $E_1^+ \cap E_3^+$ (since P_1 and P_3 commute with each other, and P_1 is the last step, P_3 the first step of block 3). Then P_4 applied to E_3^+ cannot annihilate: any $v \in E_3^+ \cap E_4^-$ would violate adjacent disjointness. □

3.5 Component 5: Final Block ($k = n-1$) and Rank Conclusion

For the final block, the same within-block argument applies. The staircase terminates with P_1 (the last step of the last block), giving $\mathrm{im}(\Pi) \subset E_1^+$.

4 The Even-Block Gap ($k \geq 6$)

4.1 Statement of the Gap

At even block $k \geq 6$, the cascade image entering block k lies in $E_1^+ \cap E_{k-1}^+$ (from the last P_1 of block $k-1$ and P_{k-1} from the first step of block $k-1$). The critical step is P_k: the operator $(1 + s_k)/2$ applied to the image after P_{k-1}.

By parabolic reduction, the relevant structure involves the S_4 subalgebra $\langle s_{k-2}, s_{k-1}, s_k \rangle$.

Proposition 9 (Gap Reduction)**.** *The even-block-k non-annihilation reduces to:*

$$(1 + s_k)(1 + s_{k-2}) \text{ is injective on } \mathcal{I} \cap E_{k-1}^+$$

where $\mathcal{I}$ is the cascade image after blocks $1, \ldots, k-2$ and P_{k-1}.

Since s_k and s_{k-2} commute ($|k - (k-2)| = 2$), the operator $Q = (1 + s_k)(1 + s_{k-2})/4 = P_k P_{k-2}$ is the projection onto $E_{k-2}^+ \cap E_k^+$.

4.2 Partial Reduction via Adjacent Disjointness

Proposition 10. *Within E_{k-1}^+:*

$$\ker(Q) \cap E_{k-1}^+ \;=\; (E_{k-2}^+ \cap E_k^-) \cap E_{k-1}^+.$$

The components $(E_{k-2}^- \cap E_k^\pm) \cap E_{k-1}^+$ vanish by adjacent disjointness.

Proof. Since s_{k-2} and s_k commute:

$$\ker(Q) = (E_{k-2}^+ \cap E_k^-) \oplus (E_{k-2}^- \cap E_k^+) \oplus (E_{k-2}^- \cap E_k^-).$$

By Lemma **??** with $i = k-2$, $i+1 = k-1$: $E_{k-2}^- \cap E_{k-1}^+ = \{0\}$. Therefore any vector in $(E_{k-2}^- \cap E_k^\pm) \cap E_{k-1}^+$ lies in $E_{k-2}^- \cap E_{k-1}^+ = \{0\}$. □

4.3 The Remaining Obstruction

Proposition **??** reduces the gap to showing:

$$\mathcal{I} \cap (E^+_{k-2} \cap E^-_k \cap E^+_{k-1}) = \{0\}.$$

In each $S_4 = \langle s_{k-2}, s_{k-1}, s_k \rangle$ irrep, the *pure* triple intersection $E^+_{k-2} \cap E^-_k \cap E^+_{k-1}$ is zero (verified by explicit computation in all five S_4 irreps). However, this does **not** imply the intersection is zero in V_λ: a vector can have components in different S_4 isotypic components that individually have zero pure triple intersection but combine to create a nonzero element of $E^+_{k-2} \cap E^-_k$ within E^+_{k-1}.

Remark 11 (Why Adjacent Disjointness Fails for $(k-2, k)$). The pair (s_{k-2}, s_k) has $|k-2-k| = 2$, so they commute: there is no S_3 subalgebra $\langle s_{k-2}, s_k \rangle$ (since they satisfy $s_{k-2}s_k = s_k s_{k-2}$, not a braid relation). The joint eigenspace $E^+_{k-2} \cap E^-_k$ can be nonzero. Adjacent disjointness applies only to *adjacent* generators.

4.4 Approaches Explored

1. Spectral separation by s_{k-3}. For the gap step, one might hope that the cascade image and the kernel of Q have disjoint s_{k-3}-spectra. Computation shows the spectra overlap (e.g., for $(4, 2, 1) \vdash 7$: kernel max $= 0.75$, image min $= 0.33$). Dead end.

2. Tight complementarity. For $k = 6$ in S_7: $\dim(\mathcal{I}) + \dim(\ker(Q) \cap E^+_1 \cap E^+_5) = \dim(E^+_1 \cap E^+_5)$ in all 15 irreps (exact complementarity). However, this fails for $k = 8$ in S_9 and does not imply transversality.

3. Hecke deformation. Under the standard isomorphism $H_n(q) \cong \mathbb{C}[S_n]$ (for $q \neq 0, -1$), the Hecke idempotent $\pi_i = (T_i + 1)/(q + 1)$ maps to $P_i = (1 + s_i)/2$, independent of q. So $\Pi_q = \Pi$ and rank is trivially q-constant. The proper Hecke seminormal form gives q-dependent eigenvalue polynomials $q^a(1 + q)^b \cdot p(q)$ with $p(1) > 0$, but proving p has no positive real roots requires the same cascade mechanisms. No shortcut.

5 Computational Verification

The theorem is verified for all $n \leq 9$ (all irreps, exact arithmetic) and for $n \leq 16$ (all irreps, floating-point with tolerance 10^{-10}).

For the specific gap step: $Q = (1 + s_{k-2})(1 + s_k)/4$ is injective on the cascade image $\mathcal{I} \cap E^+_{k-1}$ for:

- $k = 6$, S_7: all 15 irreps, minimum singular value ≥ 0.165.
- $k = 8$, S_9: all 30 irreps, minimum singular value ≥ 0.094.
- $k = 6$, S_9 (parabolic embedding): all 30 irreps.

6 What Would Close the Gap

Any ONE of the following suffices:

1. **Direct transversality:** Show $\mathcal{I} \cap (E^+_{k-2} \cap E^-_k \cap E^+_{k-1}) = \{0\}$ for all V_λ and all even $k \geq 6$.
2. **Total rank identity:** Show $\sum_\lambda \dim(V_\lambda) \cdot \operatorname{rank}(\Pi|_{V_\lambda}) = n!/2^{\lfloor n/2 \rfloor}$ algebraically. Combined with rank $\leq \dim(V^H)$ (per-irrep upper bound, itself unproved), the Frobenius squeeze gives equality.

3. **Eigenvalue positivity:** Show all roots of the Hecke eigenvalue polynomials of Π_q lie in $(-\infty, 0]$. This gives $\mathrm{rank}(\Pi_1) = \text{generic rank} = \dim(V^H)$.

4. **Guilhot–Poulain d'Andecy connection:** Their Prop 4.4 gives $\dim(e_J V_\lambda) = \dim(V_\lambda^{W_J})$ for $J = \{s_1, s_3, \ldots\}$. If the staircase product Π can be related to e_J in the Hecke algebra (e.g., $\Pi \cdot \mathbb{C}[S_n] = e_J \cdot \mathbb{C}[S_n]$ as left ideals), the theorem follows.